

Stacked dielectric resonator antenna report at 110 GHz

Bottom layer:

For the first antenna topology (Antenna1), Fig. 1 shows a ground plane with a slot, placed on a dielectric substrate of relative permittivity fed by a microstrip line that is designed and simulated. The patch dimension, slot dimension and the substrate thickness and permittivity is designed for optimum coupling.

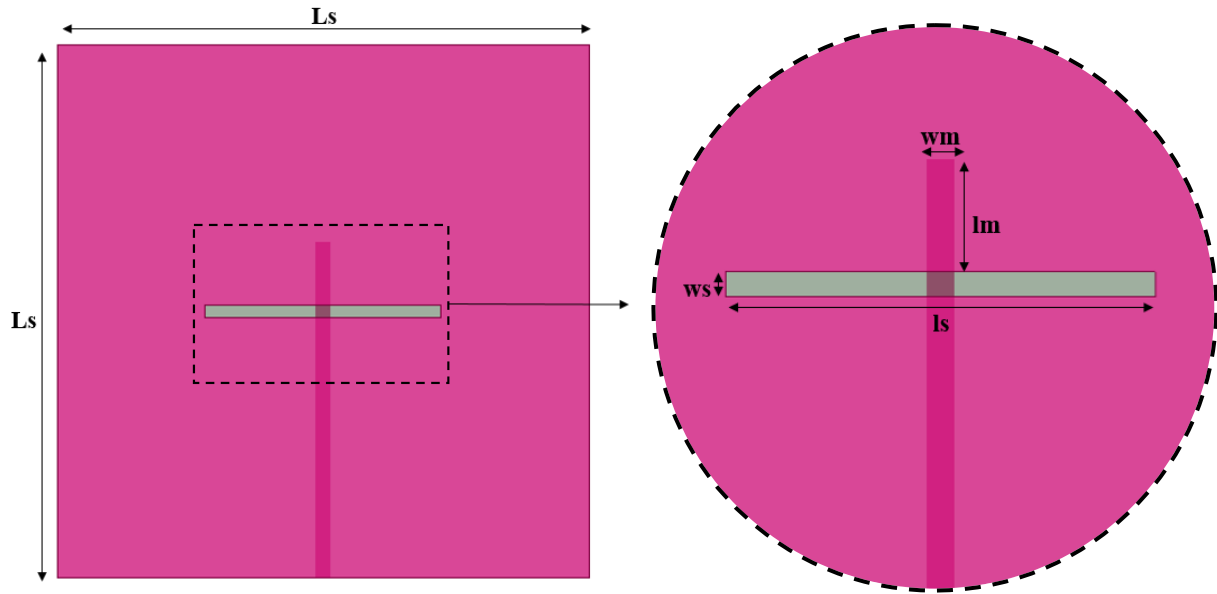


Fig.1. Base substrate with the slotted ground plane on the top and microstrip line on the bottom face

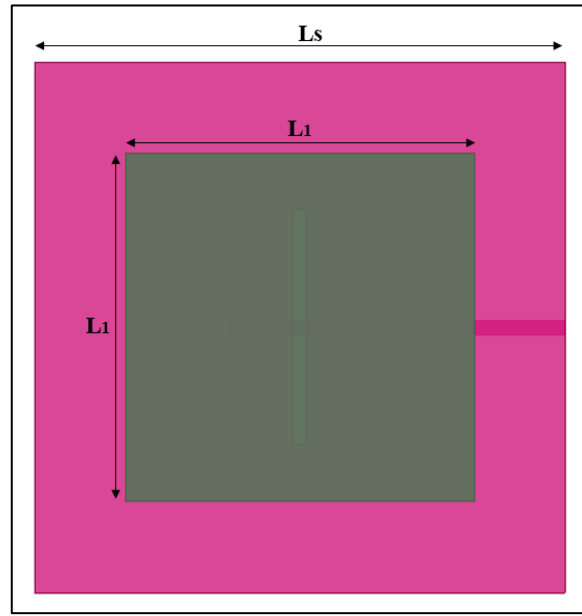
The substrate, having a relative permittivity of 3.38 has dimensions of 3.15 mm in length, 3.15 mm in width, and 0.055 mm in thickness. The rectangular slot is incorporated into the patch with a slot width of 0.081 mm, while the length is 1.395 mm, and the microstrip line width is 0.0846 mm. The micro strip line extends 1.98 mm. These dimensions are critical for the design and optimization of the antenna's performance. The dimensions are tabulated in Table 1.

Table 1: Dimensions in Fig.1

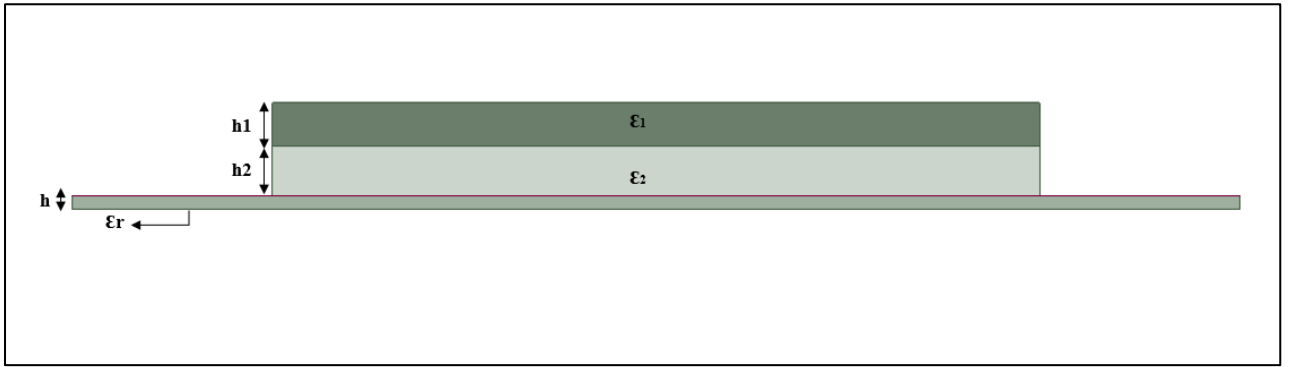
Parameters	Description	Dimensions (mm)
Ls	Length of substrate	3.15
ls	Length of slot	1.395139
ws	Width of slot	0.081
lm	Length of microstrip line	0.405
wm	Width of microstrip line	0.0846

Complete system:

For the second antenna topology (Antenna2), Fig. 2 shows the two dielectric layers stacked on top of each other and placed on top of the base substrate (Fig.1). The height of the middle substrate is 0.135mm with a relative permittivity of 2.2 and the top substrate has a height of 0.117mm with a relative permittivity of 15. Fig.2(a) shows the top view of the antenna system while Fig.2(b) shows the side view.



(a) Top view



(b) Side view

Fig. 2. Entire antenna system consisting of 3 layers.

The dimensions are tabulated in Table 2.

Table 2: Dimensions in Fig.2

Parameters	Description	Dimensions (mm)
L_1	Length of layered substrates	2.07
h_1	Height of top substrate	0.117
h_2	Height of middle substrate	0.135
h	Height of bottom substrate	0.055
ϵ_1	Relative permittivity of top layer	15
ϵ_2	Relative permittivity of middle layer	2.2
ϵ_r	Relative permittivity of bottom layer	3.38

Results:

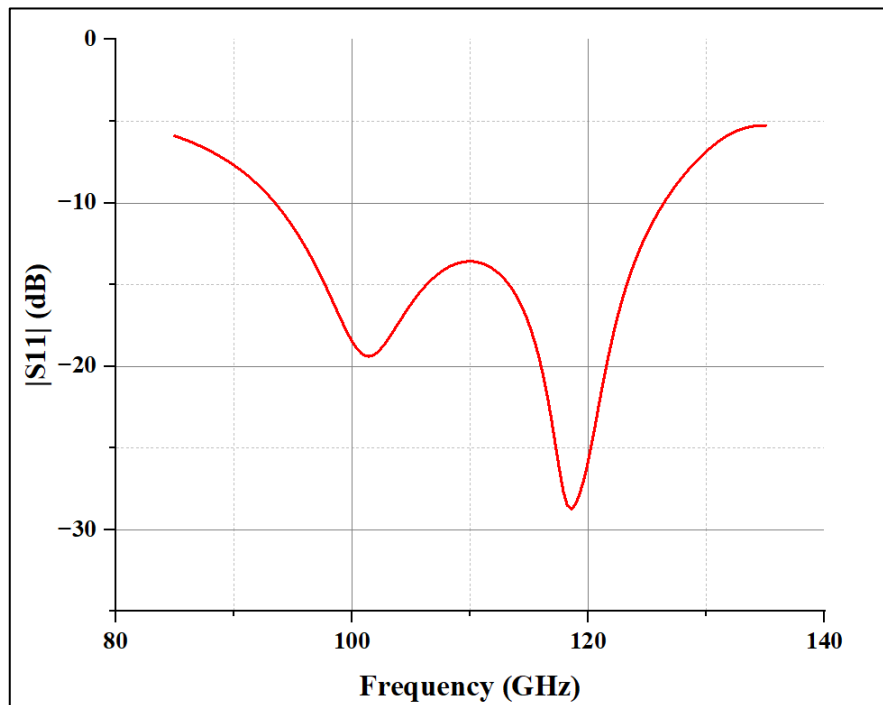


Fig. 3. Reflection coefficient vs frequency of the dielectric resonator antenna.

The resonant frequencies are obtained at **102 GHz** and **118 GHz** with good coupling characteristics. Reflection coefficients of **-19.4 dB** and **-29 dB** are achieved respectively. The reflection coefficient at 110GHz (center frequency) is observed to be -13.6 dB. The percentage bandwidth is 30%.

The gain Vs frequency curve (Fig.4) plotted for the simulation shows a gain of **9.11 dB** at **102 GHz** and a gain of **10.5 dB** at **118 GHz**.

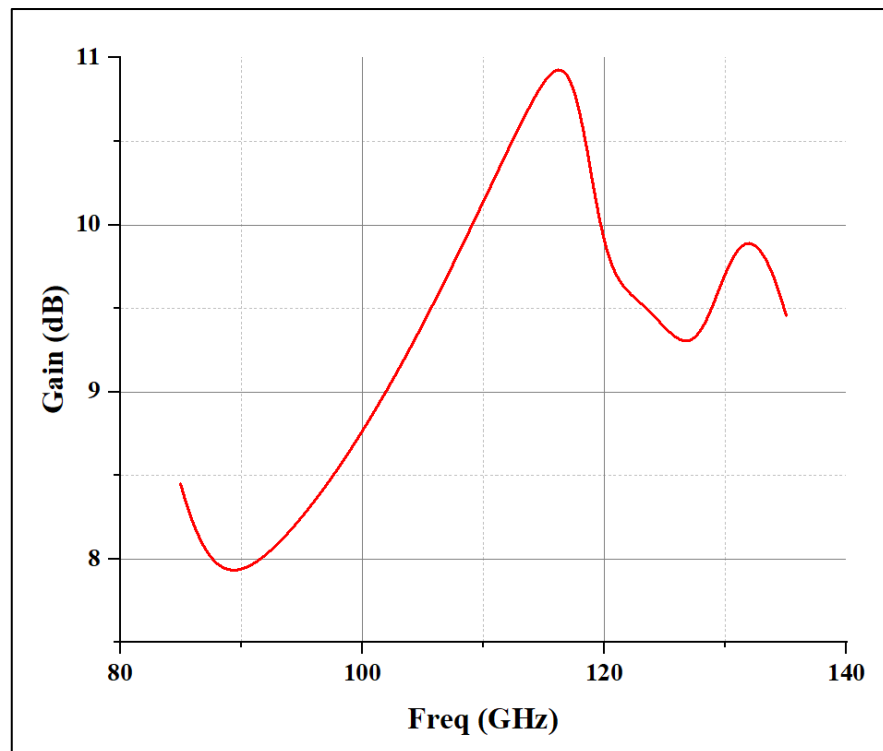


Fig. 4. Gain vs frequency of the dielectric resonator antenna.

The VSWR Vs frequency curve (Fig.4) plotted for the simulation shows a dip at 102 GHz and the lowest dip of **1.07 at 118 GHz**.

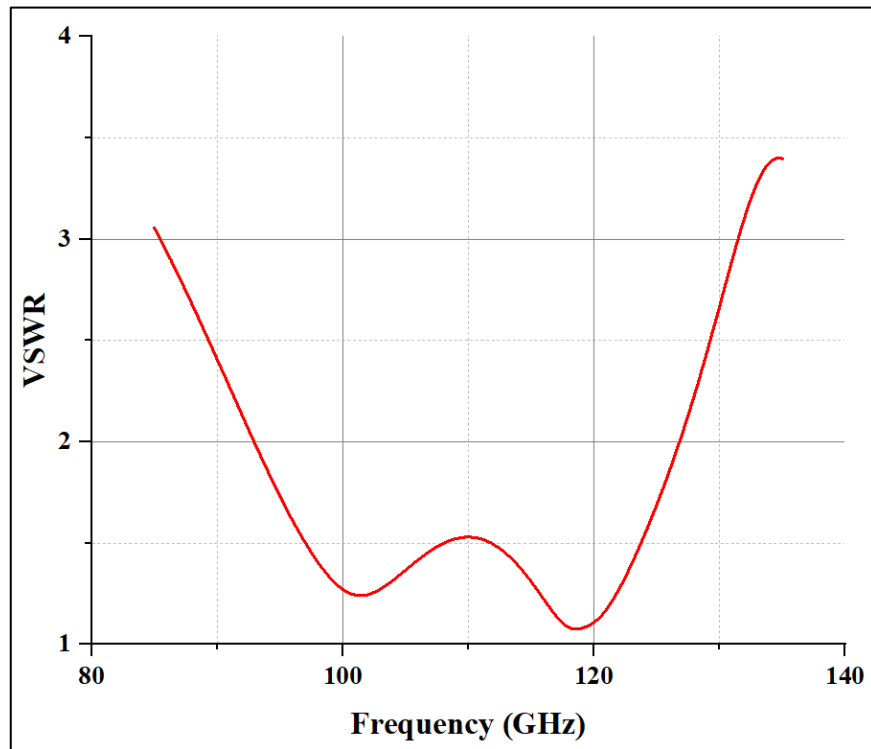
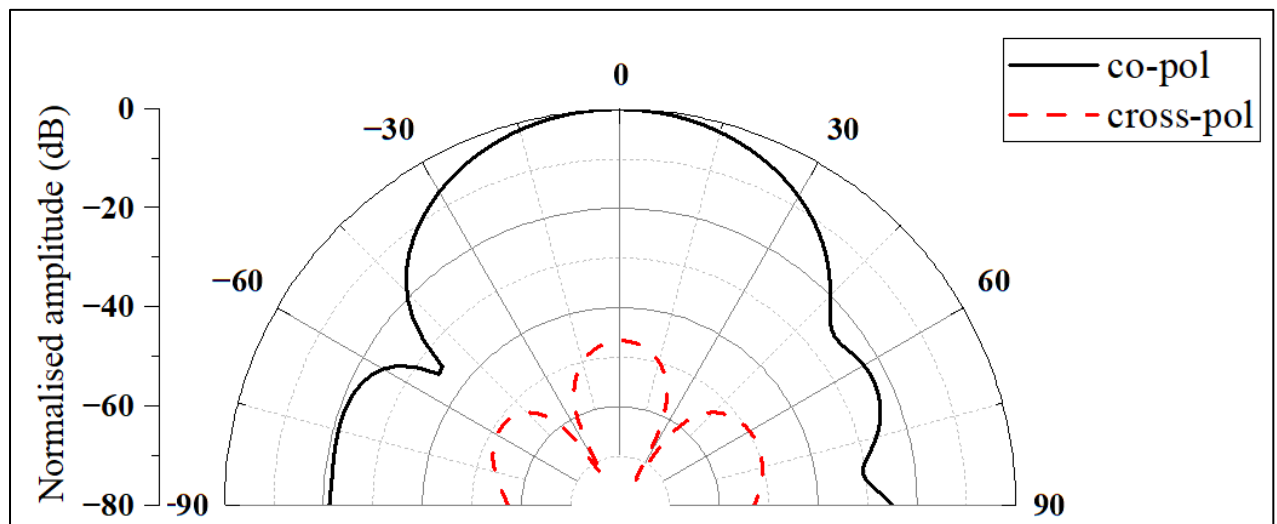
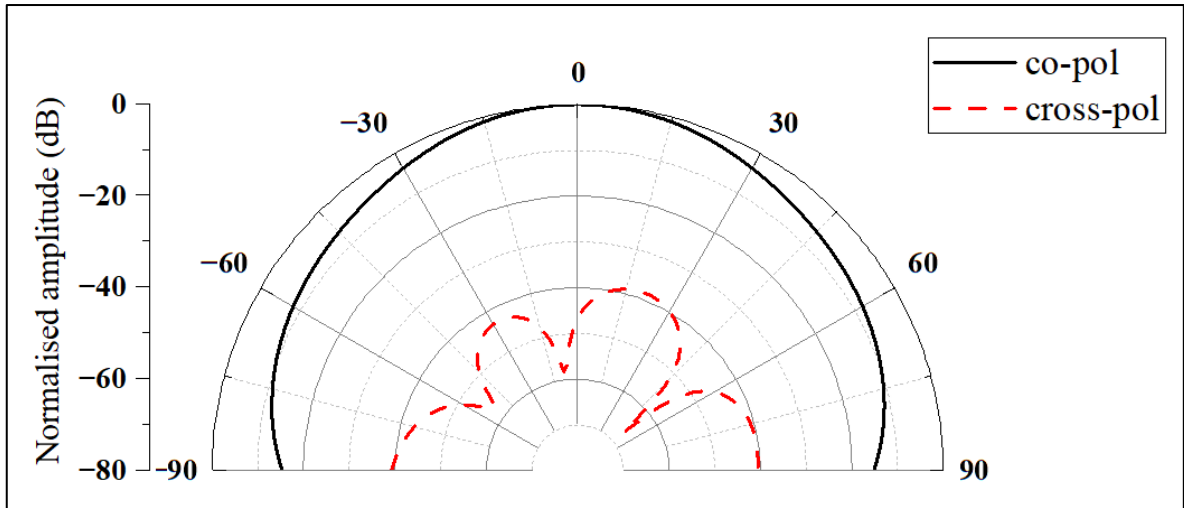


Fig. 5. VSWR vs frequency of the dielectric resonator antenna.

The radiation patterns at the resonant frequencies are shown in Fig. 6 and Fig.7 respectively.

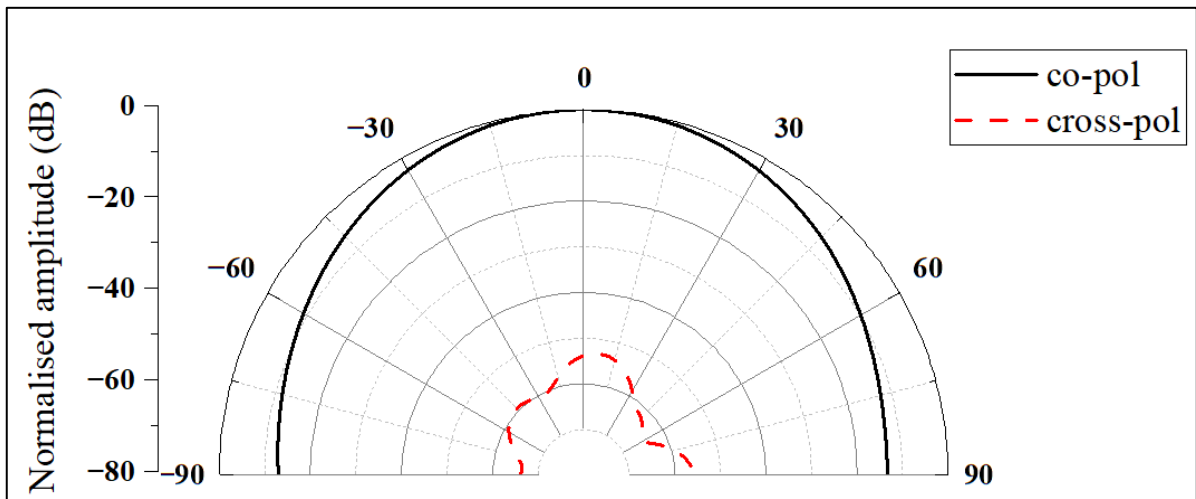


(a) $\phi = 0^\circ$

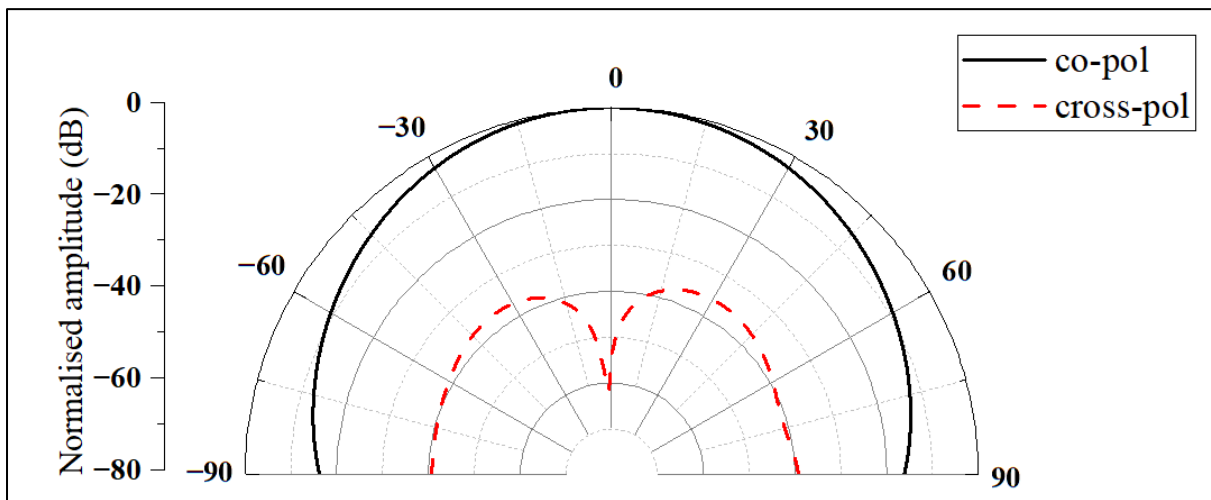


(b) $\phi = 90^\circ$

Fig. 6. Radiation pattern of the dielectric resonator antenna at 118GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$



(a) $\phi = 0^\circ$



(b) $\phi = 90^\circ$

Fig. 7. Radiation pattern of the dielectric resonator antenna at 102 GHz (a) $\phi = 0^\circ$ (b) $\phi = 90^\circ$

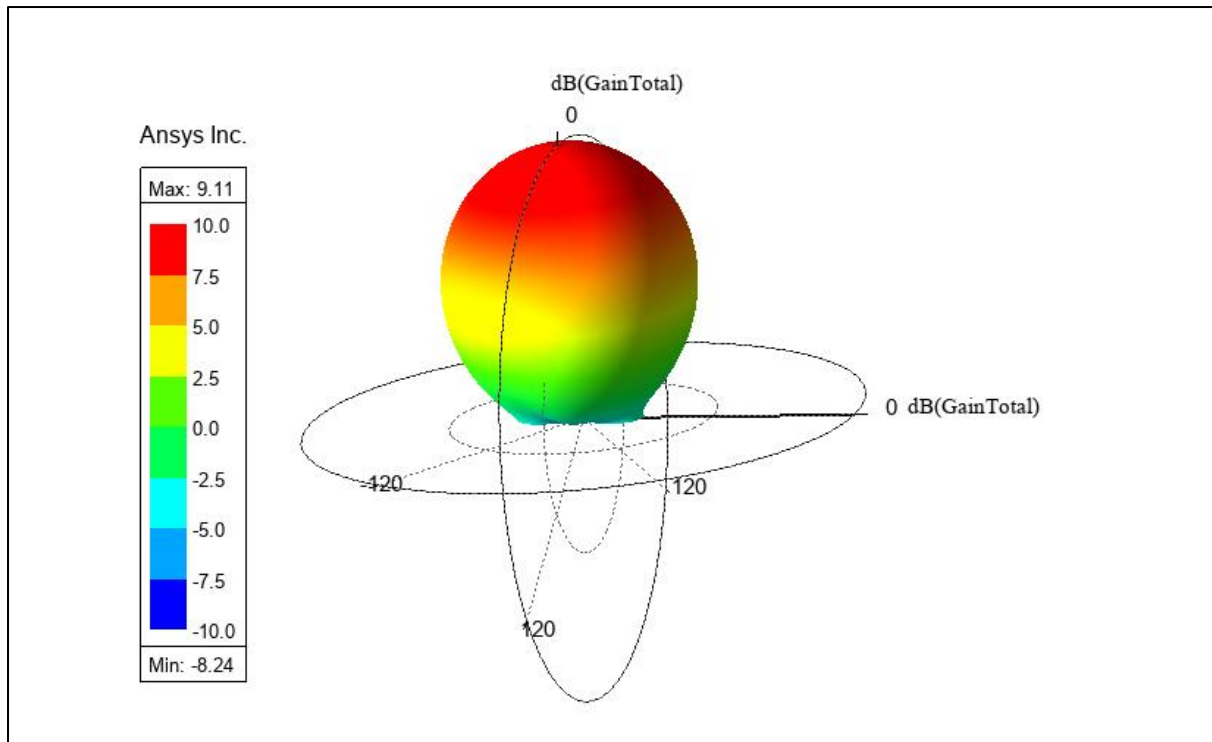


Fig. 8. 3-dimensional Radiation pattern of the dielectric resonator antenna at 102GHz

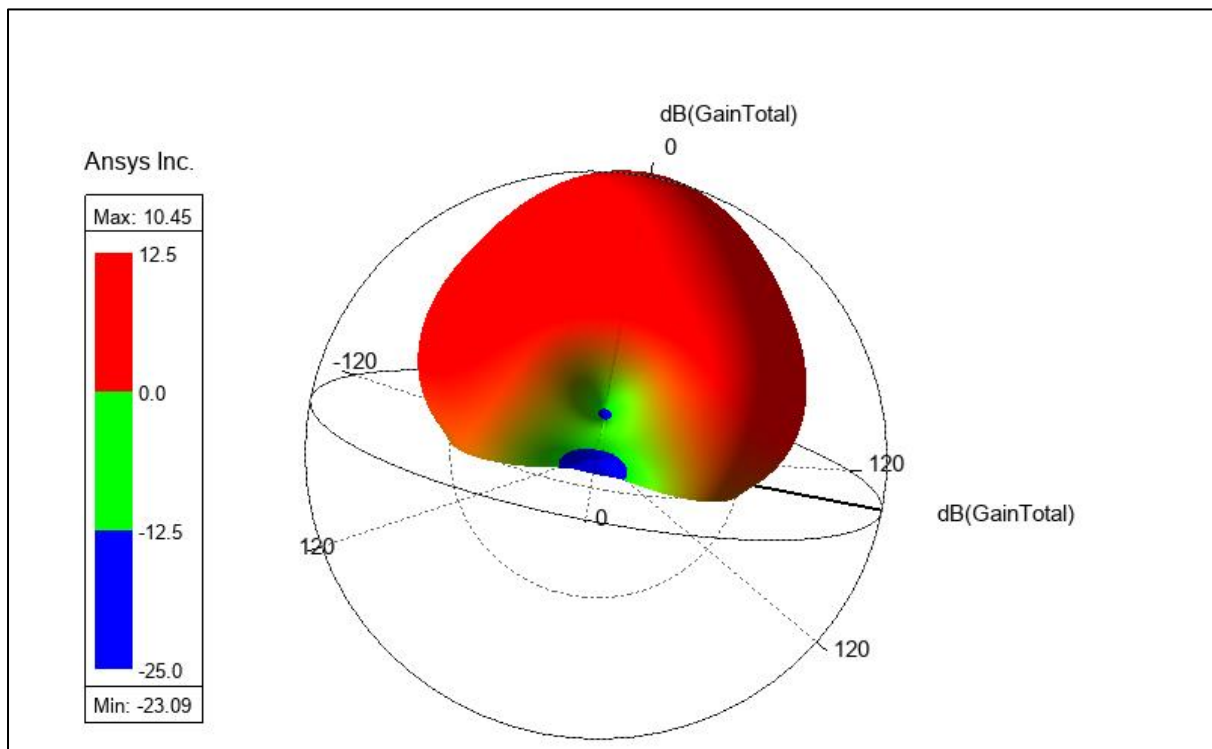


Fig. 9. 3-dimensional Radiation pattern of the dielectric resonator antenna at 118GHz

From figure 8, we can observe that the maximum gain of the stacked resonator antenna at the first resonant frequency of **102 GHz is 9.11 dB**, and from figure 9, **the gain at 118 GHz is 10.45 dB**.